SAMPLE SLIDES

Mississippi State · Ph.D. research

Power Factor Correction

ePLL-based power factor correction control with a regulated 400 V DC output, validated in real-time HIL.

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Problem & motivation

A plain diode rectifier draws current in short, distorted pulses. That gives a poor power factor and high harmonic content, wastes distribution capacity and can break grid harmonic limits such as IEC 61000-3-2.

A PFC stage shapes the input current to follow the grid voltage, but only if its control stays accurately synchronized to the grid and keeps the DC output stable as conditions change.

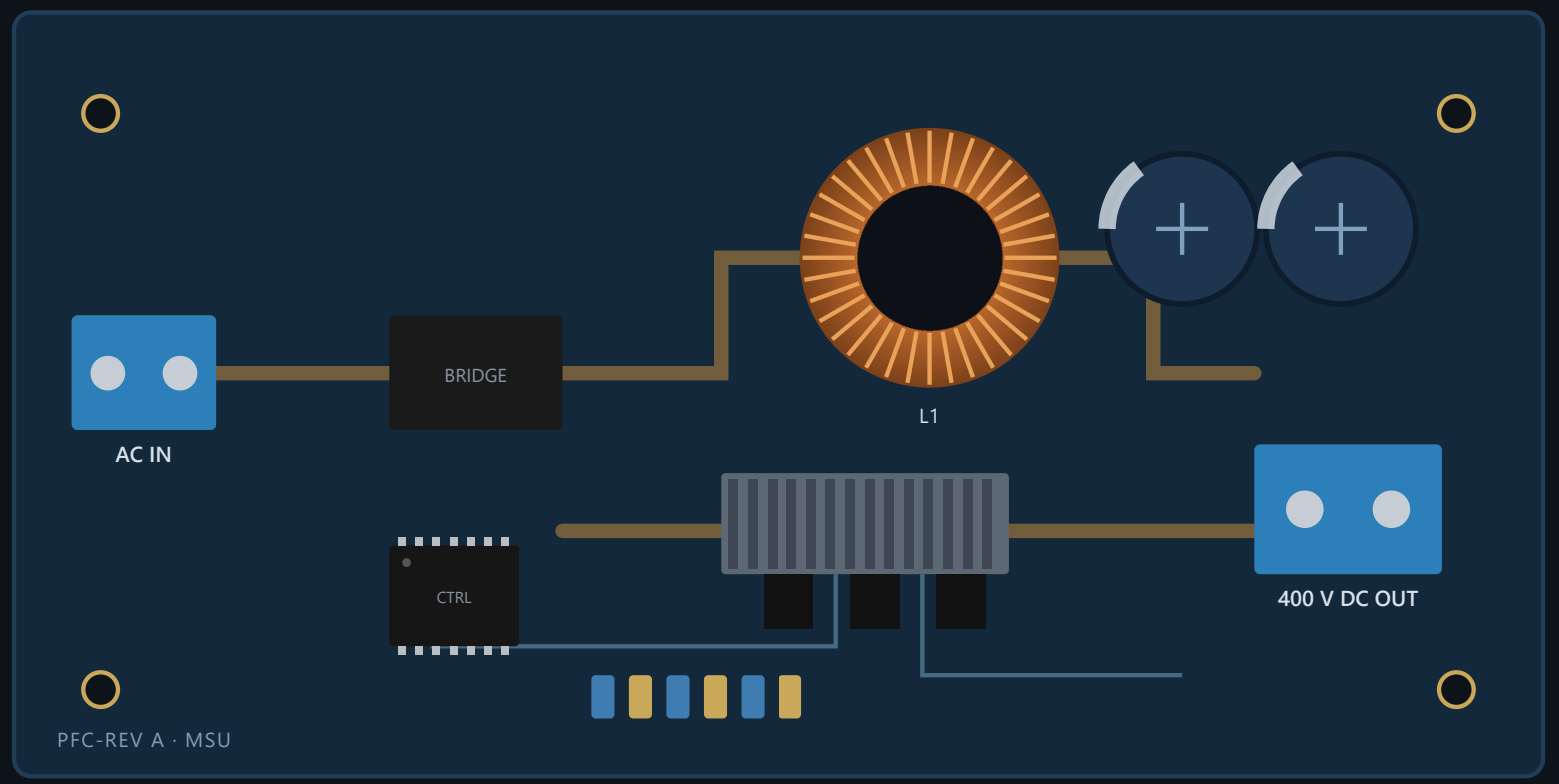
Approach

- Developed the PFC control strategy in MATLAB/Simulink with an ePLL for accurate grid synchronization
- Designed the control architecture to regulate the DC output at 400 V while improving power factor and reducing input current harmonics
- Performed real-time HIL validation on PLECS RT Box for rapid prototyping and verification
- Tested the control under dynamic operating conditions

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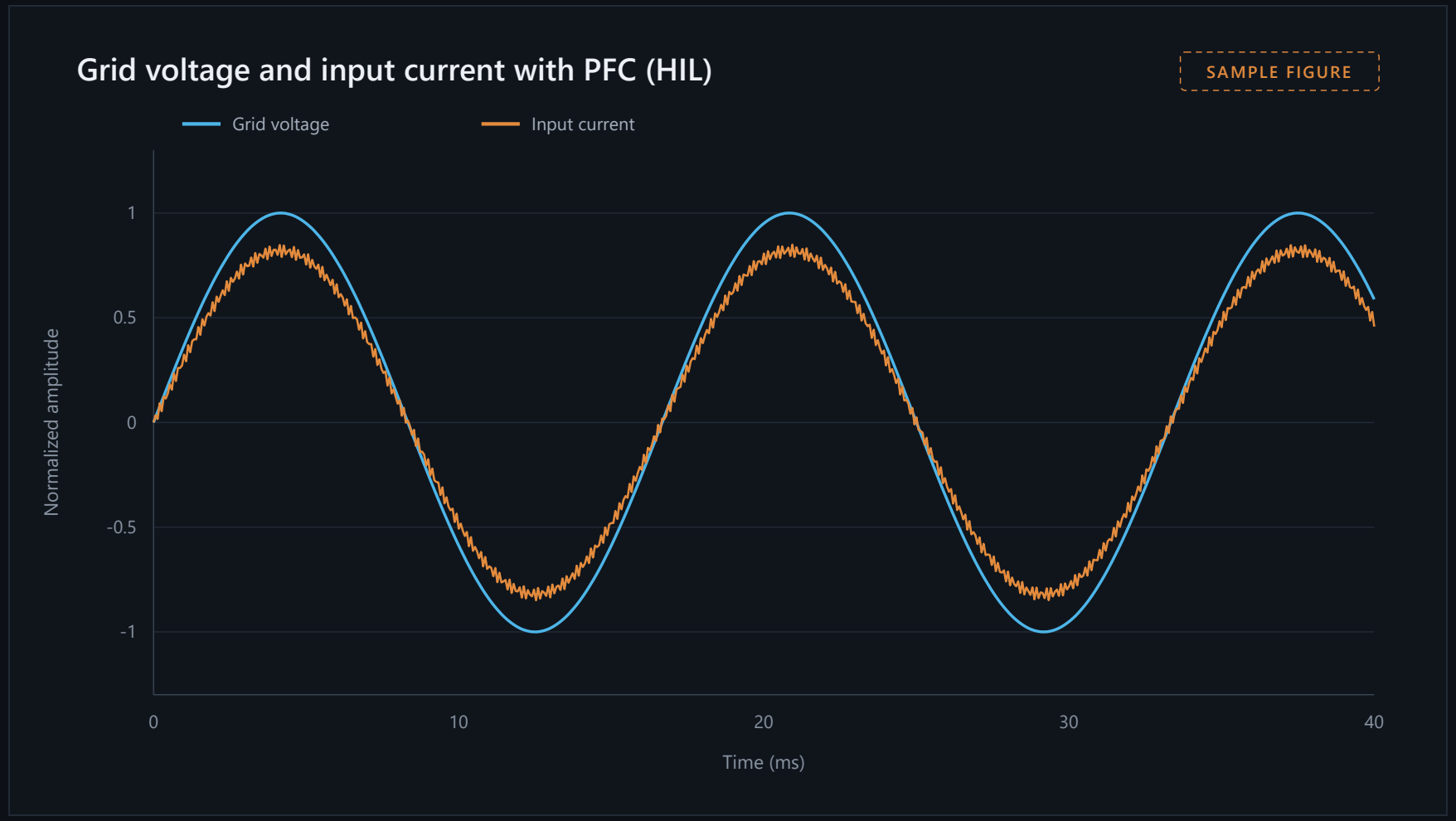
PFC board: AC input, bridge, boost inductor, switches on the heatsink, and the 400 V DC output.

SAMPLE PICTURE

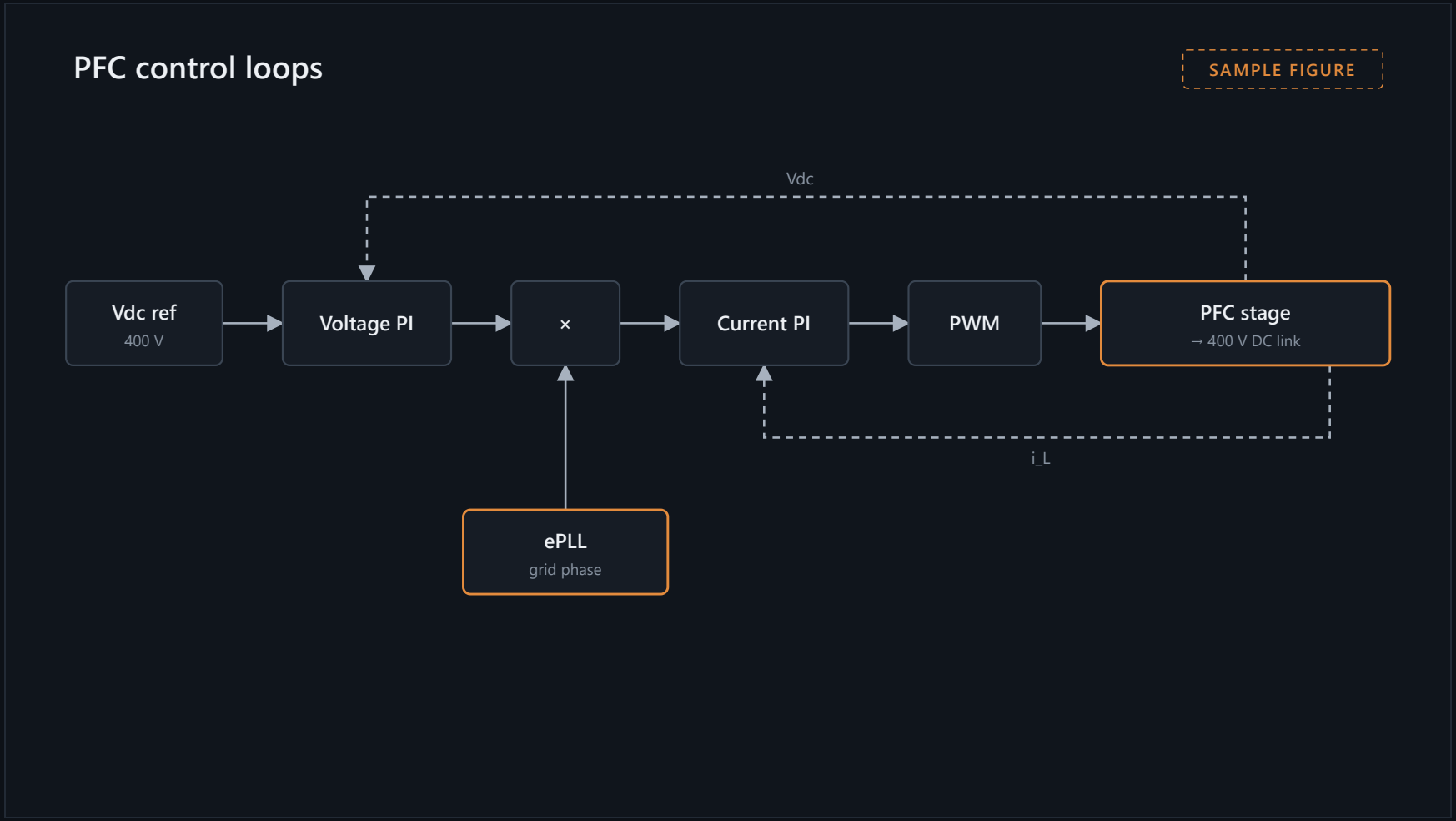


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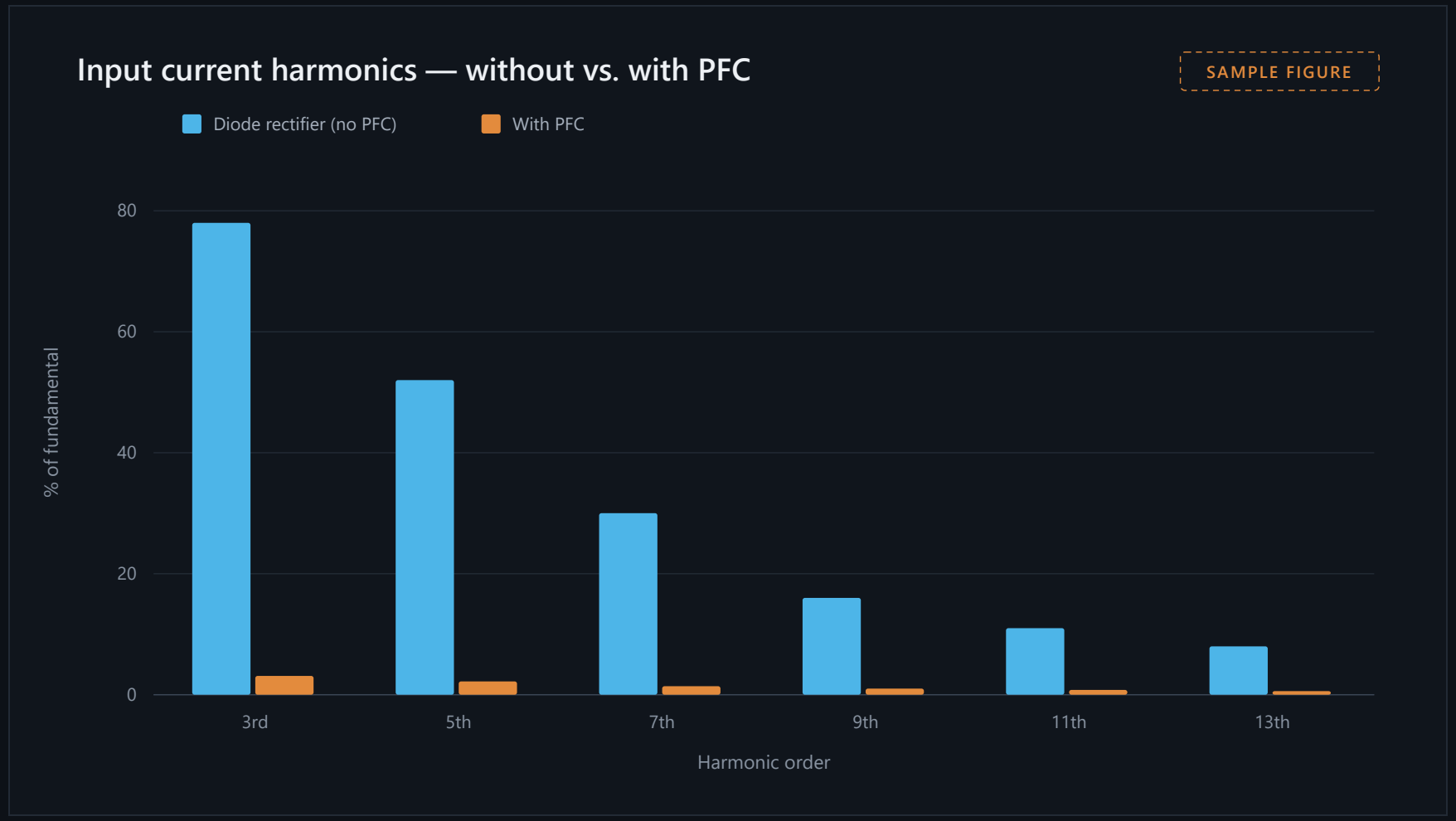
Grid voltage and input current in phase, from HIL.



Control structure: voltage loop, ePLL and current loop.



Input current harmonics with and without PFC.



Results

400 V	ePLL	0.99	3.2%
Regulated DC output	Grid synchronization	Power factor	Input current THD

- DC output regulated at 400 V
- Power factor of 0.99 and input current THD of 3.2% in HIL

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Thank you

Questions and discussion

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